

Reg No:

T376222AD224



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401

Periodical Test - II - October 2025

VII Semester

22AI702 & AI FOR ROBOTICS

Degree & Branch: B.Tech. & AD

Time: 1:30 hrs

Maximum: 50 Marks

Instructions:

1. Students should not mark/write anything on the Question Paper other than the register number.
2. Section A of the Question Paper contains questions for 15 Marks. Sections B and C contain questions for 30 Marks each.
3. Students can attempt answering any two out of three subsections in each section. The maximum mark is limited to 10 in section A and 20 in section B & C.

Q. No.	Questions	
Section - A	COURSE OUTCOME 3	MAXIMUM: 10 MARKS
A1	(i)	A delivery robot identifies door frames in a hallway for path planning. Identify the role of these detected elements. a) Irrelevant data b) Sources of error c) Random noise d) Landmarks for localization (1 Mark - [An/C, 1])
	(ii)	A mobile robot uses multiple sensors (camera, LiDAR, and wheel encoders) for localization. However, errors in measurement can accumulate and affect navigation. 1. Identify two major sources of error propagation in this scenario. (2 Marks) 2. Analyze how error propagation impacts the robot's localization and path planning. (2 Marks) (4 Marks - [An/C, 2])
A2	(i)	An indoor service robot navigates office rooms using corner features such as wall intersections and door frames. These corners act as stable reference points for localization. Predict the effect on the robot's localization accuracy if corner features are not utilized. a) Localization becomes less accurate due to loss of stable reference points. b) Localization improves since fewer features reduce processing. c) Localization remains unchanged because corners are redundant. d) Localization fails completely as no other features can be used. (1 Mark - [U/C, 1])
	(ii)	A mobile robot uses wheel encoders and LiDAR sensors for localization in an indoor environment. Over time, errors in sensor readings can accumulate, affecting navigation accuracy.

		a) Predict whether errors in wheel encoders always lead to inaccurate localization over time. (True/False) (1m) b) Evaluate the effect of sensor error propagation on the robot's navigation and path planning. (3m) (4 Marks - [An/C, 2])
3	(i)	A vision robot extracts texture patterns from floor tiles to follow a path. Predict the function of these extracted patterns a) Path guidance features b) Noise indicators c) Sensor drift causes d) Error propagation (1 Mark - [An/C, 1])
	(ii)	A warehouse robot uses cameras and LiDAR to move between shelves, avoid forklifts, and deliver packages. It processes raw sensor data to extract meaningful features for safe and efficient navigation. Statements: a) Feature extraction allows the robot to convert raw LiDAR and camera data into useful features such as corners of shelves or edges of pallets. b) The robot can still navigate accurately even if it completely skips extracting features from its sensor data. c) Extracted features help the robot localize itself in the warehouse, detect moving forklifts, and plan safe delivery paths. d) Feature extraction only adds unnecessary computational load and does not improve the robot's perception of the warehouse. 1. Identify the correct statements from the above list. (1 mark) 2. Justify the chosen statements by linking each to the robot's ability to perceive and navigate the environment. (2 mark) 3. Analyze the impact on the robot's navigation if feature extraction is skipped. (1 mark) (4 Marks - [An/C, 2])

Section - B

COURSE OUTCOME 4

MAXIMUM: 20 MARKS

(i)	A self-driving car uses LiDAR and cameras for localization. In foggy conditions, camera input becomes noisy, but LiDAR helps estimate position. Identify the principle of localization-based navigation is applied here. a) Single-sensor dependency b) Multi-sensor redundancy c) Ignoring environmental effects d) Manual override (1 Mark - [U/C, 2])
(ii)	A warehouse robot enters a new aisle. Two aisles look the same, so the robot assigns equal probability to being in either aisle. Predict the type of belief representation handles this ambiguity best. a) Single Gaussian b) Particle filter c) Dead reckoning d) GPS localization (1 Mark - [Ap/C, 1])
(iii)	A drone is flying indoors in an environment where GPS signals are not available. To determine its position, the drone cannot rely on exact coordinates. Instead, it represents its location using a probability distribution (belief). This distribution indicates how likely the drone is to be at different positions within the space. Rearrange the steps in belief representation:

	a. The Drone assumes a starting position with some confidence but still considers nearby alternatives. b. The drone gathers data from cameras, sonar, or LiDAR about walls, doors, or obstacles. c. The refined distribution gives the most likely current position, ready for the next cycle. d. The probability spreads over multiple possible positions, reflecting increased uncertainty. e. Small motion errors (motor noise, air disturbance) make its exact position less certain. f. Sensor data eliminates unlikely positions and strengthens the probability of correct ones.	(3 Marks - [An/C, 2])
(iv)	In a large shopping mall, a cleaning robot is deployed early in the morning to autonomously move through corridors, detect trash, and avoid obstacles. The robot relies mainly on sonar sensors to detect nearby walls and objects. However, while navigating: a) When passing by glass walls and shiny surfaces, the sonar beams bounce irregularly, producing false or fluctuating distance readings. This is an example of sensor noise. b) In another section of the mall, the robot enters two identical-looking corridors with the same width and layout. Its sensor data cannot distinguish between the two, and it becomes confused about its location. This is an example of aliasing. Predict how noise and aliasing affect localization in this scenario.	(5 Marks - [Ap/C, 2])
(i)	A tractor navigating a smart farm identifies rows of crops, water tanks, and farm roads with meaningful labels in its environment. Predict the most suitable map . a) Occupancy grid map b) Metric map c) Semantic map d) Topological map	(1 Mark - [Ap/C, 1])
(ii)	A rescue robot explores a collapsed mine, creating a real-time map of tunnels. Identify the reason for autonomous map building essential here. a) GPS is not available underground b) It reduces sensor noise c) It eliminates the need for cameras d) It avoids charging batteries	(1 Mark - [U/C, 2])
(iii)	A mall security robot patrols multiple floors of a large shopping center. It needs to monitor entrances, escalators, and corridors efficiently while coordinating with other robots. Predict the statements from the following which are correctly describe how map representation supports the robot's operations. Options: 1. Maps allow the robot to plan patrol routes efficiently. 2. Maps help avoid collisions with walls, escalators, or people. 3. The robot can perform tasks without any sensors using only maps. 4. Map representation enables coordination with other security robots. 5. Using maps slows down the robot's response to dynamic events. 6. Maps help the robot quickly locate areas that need urgent attention	(4 Marks - [U/C, 2])
(iv)	A delivery robot is deployed in a large shopping mall to help guide customers to various shops, restaurants, and food courts. The environment is complex, containing open atriums, narrow corridors, escalators, and dynamic areas with heavy human movement.	

	Predict which map representation technique the robot would use to ensure efficient navigation, and justify your reasoning with any three supporting points. (4 Marks - [Ap/C, 2])
(i)	A drone patrols a forest to detect illegal logging. GPS is weak under dense trees. The drone uses Visual SLAM to localize itself while navigating. Represent the main advantage of SLAM here. - It ignores localization errors - It builds and updates maps while localizing - It relies only on GPS - It prevents the need for sensors (1 Mark - [U/C, 2])
(ii)	A food delivery robot uses GPS + IMU + cameras to update its position while navigating streets. If it relies only on GPS, localization is inaccurate in tunnels. Identify the approach improves navigation accuracy. - Dead reckoning only - Sensor fusion for localization - Ignoring localization and using fixed routes - Using a single camera feed (1 Mark - [U/C, 2])
(iii)	An indoor drone explores an office floor without GPS support. Instead of assuming a single fixed position, it maintains a probability distribution of its possible locations. With each movement, uncertainty increases, but sensor inputs like wall distances and camera readings refine its belief about where it is. From the above scenario, Predict how a belief representation helps the drone improve localization. Give three specific benefits. (3 Marks - [An/C, 2])
(iv)	At a sports stadium, a food delivery robot moves between the kitchen and food stands. To avoid colliding with fans and to deliver food correctly, the robot must know its location continuously. Localization plays a major role in ensuring safe and efficient navigation. Read the following statements about the role of localization in this scenario and then map them with the correct short options. Statements: - When the robot integrates localization with mapping, it can plan efficient delivery routes and ensure smooth movement across the stadium corridors. - If the robot does not perform localization, its navigation decisions will turn into trial-and-error, increasing the chances of missing stalls and wasting time. - By continuously updating its position, the robot can detect moving fans in real time and avoid potential collisions. - With localization, the robot can re-calculate its path dynamically when a corridor is blocked by crowds or trolleys, ensuring on-time delivery. - If the robot only depends on localization, it can completely ignore obstacle detection, since position information alone will keep it safe. Options - Continuous updates prevent collisions. - Dynamic re-routing is supported. - Mapping combines with planning. - Navigation becomes guesswork. - Obstacle detection is unnecessary. (5 Marks - [U/C, 2])

(i)	A robot explores a building with loops and dead ends, often revisiting the same areas. Find the method that improves its efficiency by preventing repeated exploration of non-productive paths. - Hill climbing for fast elevation gain - Tabu search by storing explored paths - Greedy search to reach the goal early - DFS with no backtracking (1 Mark - [Ap/C, 2])
(ii)	A robot explores a cave with no prior map available. It must build a map and navigate simultaneously. Predict a method that is suitable for the above scenario. - SLAM - PID control - BFS with known map - Kalman filtering only (1 Mark - [Ap/C, 2])
(iii)	A robot is deployed in a smart hospital to autonomously deliver medicines from the pharmacy to various patient rooms and wards, while navigating hallways, avoiding people, and handling dynamic obstacles (like moving stretchers or cleaning staff). Rearrange the following steps in the correct order to show how a hospital robot handles dynamic obstacles like moving stretchers or staff during navigation. (3 mark) - The robot reduces speed, keeps a safe distance, or waits when near humans to ensure safety. - AI models track the direction and speed of moving objects to predict their motion. - The robot senses its environment in real-time using LIDAR, cameras, or other sensors. - When the local path is blocked, the robot updates its global route to continue its task. - The robot identifies dynamic obstacles based on proximity and movement patterns. - It uses the Dynamic Window Approach (DWA) to adjust its path in real-time. (3 Marks - [Ap/C, 2])
(iv)	A warehouse robot uses sensor-based navigation to reach a designated docking station. As it approaches the goal, the path ahead remains clear of shelves and human workers. The robot steadily slows down as it nears the docking point, without making abrupt maneuvers or corrections, indicating that no new obstacles are influencing its path at that moment. A mobile robot navigates using potential fields, where the total potential is defined as: $U(q) = \frac{1}{2} k_{att} \ q - q_{goal}\ ^2 + \frac{1}{2} k_{rep} \left(\frac{1}{\rho(q)} - \frac{1}{\rho_0} \right)^2 \quad \text{if } \rho(q) \leq \rho_0$ Assuming the robot is near the goal position and there are no nearby obstacles (i.e., $\ q - q_{goal}\ \rightarrow 0$ and $\rho(q) > \rho_0$). - Find the robot's motion behavior at this state and justify the answer based on the potential field model. (4 mark) - Predict how does a net force of zero imply about the robot's state in a potential field. (1 mark) (5 Marks - [Ap/C, 3])
(i)	A robot fleet collaborates to deliver items across a factory floor. Identify the navigation architecture that allows coordinated decision-making across multiple robots. - Isolated systems with onboard-only decisions - Reactive agents operating without communication - Distributed system with shared environment models - Standalone planners with random movement control (1 Mark - [U/C, 2])

(ii)	A delivery robot receives conflicting readings from LIDAR and ultrasonic sensors. Find the approach the robot should implement to ensure reliable obstacle detection. - Use sensor fusion techniques to combine readings intelligently - Follow a default route regardless of conflicting information - Deactivate one sensor to avoid processing conflicts in data - Pause all actions and alert the user for manual override of controls (1 Mark - [Ap/C, 2])
(iii)	A warehouse delivery robot is moving toward a packing station when it detects a stack of boxes 2.5 m ahead, within its safety influence range of 3 m. Using the repulsive potential equation, its onboard controller calculates the "energy hill" created by the boxes — the closer it gets, the steeper the hill. This artificial energy pushes the robot to choose a path around the boxes instead of plowing straight into them. A robot is at $q=(2,2)$, an obstacle center is at $q_{obs}=(4,4)$, $k_{rep}=100$, and $p_0=3$ m. Find repulsive potential energy. Hint:  (3 Marks - [Ap/C, 3])
(iv)	Inside a warehouse, an autonomous robot follows a pre-planned path to reach a storage zone. As it moves, it encounters a newly placed shelf in the corridor that was previously clear on its map. The robot stops in front of the shelf, unable to continue toward its destination. Predict the causes of a warehouse robot using deliberative navigation to fail in dynamic conditions, and find the strategy that could make it more adaptable. (5 Marks - [An/C, 2])
(i)	A delivery robot encounters unexpected construction on its pre-planned path. Predict how the robot should continue its task without significant delay. - Increase speed - Wait for human input - Turn off sensors - Recompute the path based on updated map (1 Mark - [Ap/C, 2])
(ii)	A service robot moves toward an elevator, but it sees that the elevator is currently out of order. Refer the scenario, predict how the robot should adjust its plan in real time. - Wait for maintenance - Attempt to use the elevator anyway - Take an alternative route using stairs or another elevator - End the current task (1 Mark - [Ap/C, 2])
(iii)	In a fire rescue robot, DWA is best categorized as a real-time, reactive, local path planning and obstacle avoidance technique, critical for safe and adaptive movement in unpredictable and hazardous environments. Identify the type and functions of local planning algorithm suitable for the above scenario. (2 Marks - [Ap/C, 2])

A warehouse robot starts at the charging station A (0,0) and needs to pick up an item from Rack 7 at location (5,5) and deliver it to Packing Station B at (10,2).

The warehouse is represented as a 10x10 grid, where:

0 represents a free cell

1 represents a static obstacle (rack or wall)

A moving forklift crosses the path at location (6,5) during execution.

Predict how does the robot will plan the optimal path from (0,0) to (5,5) using A- star algorithm (A*).

(6 Marks - [Ap/C, 3])

*** End of Question Paper ***